

Simple Linear Regression

Inference for the Slope

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Roadmap

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1. From Estimates to Inference

1.1 Where Lecture 6 Left Off

Lecture 6 produced **point estimates**:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Plug in the data, get two numbers. Done?

Not yet. Two questions remain unanswered:

- **How precise** is $\hat{\beta}_1$? If we drew a fresh sample, would it land near the same value, or somewhere very different?
- **Is β_1 even nonzero?** I.e. does X help predict Y at all, or is the slope we see just noise?

These are inference questions, and they're the *same* questions Lecture 4 answered about the population mean μ .

1.2 The Plan: Port the L3–L4 Toolkit

L3 and L4 built an inference machine for the sample mean:

estimator $\bar{X}$ $\rightarrow$ standard error $SE(\bar{X})$ $\rightarrow$ CI $\rightarrow$ hypothesis test.

L7's plan: **port that same machine to the slope**. Every step has a direct parallel.

Inference on the mean (L3–L4)	Inference on the slope (L7)
Parameter μ	Parameter β_1
Estimator $\bar{X}$	Estimator $\hat{\beta}_1$
Standard error $SE(\bar{X})$	Standard error $SE(\hat{\beta}_1)$
CI for μ	CI for β_1
Test $H_0 : \mu = \mu_0$	Test $H_0 : \beta_1 = 0$

The only thing that changes is the standard-error *formula*. The CI and HT recipes are identical in shape.

2. Recall: Inference on the Sample Mean

2.1 The Setup

An i.i.d. random sample $X_1, X_2, \dots, X_n$ from a population with mean μ and variance σ^2 . The estimator of μ is the sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Notation reminder. $\bar{X}$ is a function of the data,
 $\bar{X} : (X_1, \dots, X_n) \mapsto \frac{1}{n} \sum X_i$. Different samples give different values of $\bar{X}$
— so $\bar{X}$ itself is a random variable with its own distribution.

2.2 The Sampling Distribution (CLT Recall)

By the Central Limit Theorem (Lecture 3 §6), for n large ($n \geq 30$),

$$\bar{X} \stackrel{\text{approx}}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right).$$

No Normality required of the population. The CLT delivers the Normal shape automatically, just by averaging.

Two consequences:

- $\bar{X}$ is centered on μ (no systematic error).
- Its spread shrinks like $1/\sqrt{n}$ as n grows.

2.3 The Standard Error

Standard error of the sample mean

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

If σ is unknown, plug in the sample standard deviation and use $\hat{SE}(\bar{X}) = s/\sqrt{n}$ (valid for $n \geq 60$).

Intuition. The standard error has two moving parts:

- More population noise $\sigma \Rightarrow$ larger SE.
- More data $n \Rightarrow$ smaller SE ($\propto 1/\sqrt{n}$).

2.4 Two Properties of an Estimator

Two ways to ask “is this a good estimator?”

Definition: unbiased

An estimator $\hat{\theta}$ of a parameter θ is **unbiased** if

$$\mathbb{E}[\hat{\theta}] = \theta.$$

On average across samples, $\hat{\theta}$ hits the parameter exactly.

Definition: consistent

A sequence of estimators $\hat{\theta}_n$ (one for each sample size n) is **consistent** for θ if

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \text{as } n \rightarrow \infty.$$

With enough data, $\hat{\theta}_n$ gets arbitrarily close to θ with probability close to one.

The sample mean is both: $\mathbb{E}[\bar{X}] = \mu$ (unbiased) and $\bar{X} \xrightarrow{P} \mu$ (LLN, consistent) — proven in Lecture 3 §3 and §5.

2.5 CI and HT for μ in One Slide

Confidence interval

$$\bar{X} \pm z_{\alpha/2} \cdot \hat{SE}(\bar{X})$$

Critical values: $z_{\alpha/2} = 1.645$ (90%), 1.96 (95%), 2.576 (99%).

Hypothesis test of $H_0 : \mu = \mu_0$

Test statistic

$$Z = \frac{\bar{X} - \mu_0}{\hat{SE}(\bar{X})} \stackrel{H_0}{\approx} N(0, 1).$$

Reject H_0 (two-sided) if $|Z| > z_{\alpha/2}$, equivalently if $p \leq \alpha$ where $p = 2\mathbb{P}(Z > |Z_{\text{obs}}|)$.

Recipe details: Lecture 4 §4 (CI) and §7 (HT). The next section reuses these formulas with $\bar{X} \rightarrow \hat{\beta}_1$.

3. The Same Story for OLS

3.1 The Model (Recall + One New Assumption)

Lecture 6 stated the model

$$Y = \beta_0 + \beta_1 X + \epsilon, \quad \mathbb{E}[\epsilon \mid X] = 0.$$

Across the sample, (X_i, Y_i) are independent. Within an observation, $\mathbb{E}[\epsilon_i \mid X_i] = 0$: the error has no systematic direction at any predictor value.

For inference we add a separate assumption on the error spread:

New assumption: homoscedasticity

The error variance is the same at every value of X :

$$\text{Var}(\epsilon \mid X = x) = \sigma^2 \quad \text{for all } x.$$

Here σ^2 is an unknown number controlling the noise spread. Software estimates it when reporting standard errors in L7 and L9.

3.2 The Parallel, Filled In

Replace “population mean” with “true slope” everywhere:

Sample mean	OLS slope
Parameter μ	Parameter β_1
Estimator $\bar{X}$	Estimator $\hat{\beta}_1$
$\mathbb{E}[\bar{X}] = \mu$	$\mathbb{E}[\hat{\beta}_1] = \beta_1$
$\bar{X} \xrightarrow{P} \mu$	$\hat{\beta}_1 \xrightarrow{P} \beta_1$
$\text{SE}(\bar{X}) = \sigma/\sqrt{n}$	$\text{SE}(\hat{\beta}_1) = ?$ (§4)
$\bar{X} \approx N(\mu, \text{SE}^2)$	$\hat{\beta}_1 \approx N(\beta_1, \text{SE}^2)$

The four ingredients we need: unbiasedness, consistency, sampling distribution, and a standard-error formula. The first three are on the next two slides; the SE formula gets its own section.

3.3 Unbiasedness of OLS

Key Result — OLS is unbiased

Under the model in §3.1,

$$\mathbb{E}[\hat{\beta}_1] = \beta_1, \quad \mathbb{E}[\hat{\beta}_0] = \beta_0.$$

Reading. Across many samples, $\hat{\beta}_1$ neither systematically over- nor under-shoots the truth β_1 . The result can be expressed compactly with the matrix machinery introduced in Lecture 8.

3.4 Consistency of OLS

Key Result — OLS is consistent

As $n \rightarrow \infty$,

$$\hat{\beta}_1 \xrightarrow{p} \beta_1, \quad \hat{\beta}_0 \xrightarrow{p} \beta_0.$$

Lecture 6 §4.3–§4.5 already did most of this work. We saw

$$\hat{\beta}_1 \xrightarrow{p} \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

by applying the LLN to numerator and denominator separately. Under the model, $\text{Cov}(X, Y) = \beta_1 \text{Var}(X)$, so the ratio equals β_1 .

Reading. With enough data, $\hat{\beta}_1$ is very close to the true slope — consistency is the long-run guarantee that the OLS recipe “works.”

3.5 Sampling Distribution of $\hat{\beta}_1$

Key Result — Large-sample Normal

For $n \geq 60$,

$$\hat{\beta}_1 \stackrel{\text{approx}}{\sim} N(\beta_1, \text{SE}(\hat{\beta}_1)^2).$$

Same shape as $\bar{X}$. Why we believe it:

- $\hat{\beta}_1$ combines information from all the observations.
- Because the pairs (X_i, Y_i) are independent across i , the CLT gives the same large-sample Normal logic that gave $\bar{X}$ its Normal shape (Lecture 3 §6).
- The mean of the sampling distribution is β_1 (unbiasedness, §3.3).
- The variance is $\text{SE}(\hat{\beta}_1)^2$, defined and discussed in §4.

The big payoff. Once we have this sampling distribution, the Lecture 4 pivot machinery applies *unchanged*. The CI and HT will just be the L4 formulas with $\hat{\beta}_1$ in place of $\bar{X}$.

4. The Standard Error

4.1 Definition and How We Compute It

Definition — standard error

The **standard error** of $\hat{\beta}_1$ is the standard deviation of its sampling distribution,

$$SE(\hat{\beta}_1) = \sqrt{\text{Var}(\hat{\beta}_1)}.$$

It measures the typical distance between $\hat{\beta}_1$ and the true β_1 across hypothetical resamples. Same definition for $SE(\hat{\beta}_0)$.

How we compute it in this course. We let the computer do the arithmetic. Every statistical package reports $\hat{SE}(\hat{\beta}_1)$ and $\hat{SE}(\hat{\beta}_0)$ in the *Std. error* column of the regression output, right next to the coefficient estimates.

What to retain. The exact formula is not a hand-computation target in this course. Software reports the estimated standard error for every OLS coefficient; we interpret and use those reported values.

4.2 Three Levers Behind the Standard Error

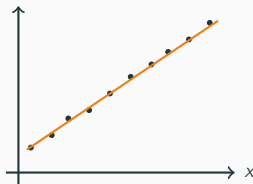
Even without the formula, three features of the data make $\hat{\beta}_1$ more precise (smaller SE):

1. **Smaller error variance σ^2** — less noise around the line means each data point pins the slope more tightly.
2. **Larger sample size n** — more observations always tighten the estimate, just like $SE(\bar{X}) \propto 1/\sqrt{n}$ in Lecture 3.
3. **Wider spread of X** — a longer “lever arm” lets the slope be identified more sharply.

Picture. Same noise, same n , different X spreads:



Narrow X spread
 $\Rightarrow$ slope poorly pinned



Wide X spread
 $\Rightarrow$ slope tightly pinned

5. Confidence Intervals for the Slope

5.1 The Pivot — Mirroring L4 §4.3

We mirror Lecture 4 §4.3, with $\hat{\beta}_1$ playing the role of $\bar{X}$.

Step 1: standardized slope is approximately Normal. For $n \geq 60$, the sampling distribution from §3.5 gives

$$\frac{\hat{\beta}_1 - \beta_1}{\widehat{SE}(\hat{\beta}_1)} \underset{\text{approx}}{\sim} N(0, 1).$$

Step 2: use the definition of $z_{\alpha/2}$.

$$\mathbb{P}\left(-z_{\alpha/2} \leq \frac{\hat{\beta}_1 - \beta_1}{\widehat{SE}(\hat{\beta}_1)} \leq z_{\alpha/2}\right) \approx 1 - \alpha.$$

5.1 The Pivot — Mirroring L4 §4.3 2

Step 3: multiply through by $\hat{SE}(\hat{\beta}_1) > 0$.

$$\mathbb{P}\left(-z_{\alpha/2} \hat{SE} \leq \hat{\beta}_1 - \beta_1 \leq z_{\alpha/2} \hat{SE}\right) \approx 1 - \alpha.$$

Step 4: subtract $\hat{\beta}_1$, then multiply by -1 . Just like L4 §4.3 Steps 5–6:

$$\mathbb{P}\left(\hat{\beta}_1 - z_{\alpha/2} \hat{SE} \leq \beta_1 \leq \hat{\beta}_1 + z_{\alpha/2} \hat{SE}\right) \approx 1 - \alpha.$$

Conclusion

The interval $\hat{\beta}_1 \pm z_{\alpha/2} \hat{SE}(\hat{\beta}_1)$ contains the true β_1 with approximate probability $1 - \alpha$.

5.2 The Formula

Key Result — CI for β_1

$$\hat{\beta}_1 \pm z_{\alpha/2} \cdot \hat{SE}(\hat{\beta}_1)$$

Critical values (same table as L4):

Confidence	α	$z_{\alpha/2}$
90%	0.10	1.645
95%	0.05	1.96
99%	0.01	2.576

Same formula for the intercept, with $\hat{SE}(\hat{\beta}_0)$ in place of $\hat{SE}(\hat{\beta}_1)$.

5.3 Recipe

How to build a CI for β_1

1. Run OLS on the data. Read off $\hat{\beta}_1$ and $\hat{SE}(\hat{\beta}_1)$ from the output.
2. Pick a confidence level $1 - \alpha$; look up $z_{\alpha/2}$.
3. Compute the margin of error: $ME = z_{\alpha/2} \cdot \hat{SE}(\hat{\beta}_1)$.
4. Report the CI: $[\hat{\beta}_1 - ME, \hat{\beta}_1 + ME]$.
5. Interpret in the units of the problem.

Identical in shape to the L4 §4.4 recipe for μ . Two examples follow.

5.4 Example 1: Sales on TV Advertising

Setup. A regional retailer regresses unit sales (in thousands) on TV ad budget (in thousands of dollars) across $n = 200$ markets. The OLS output is:

	Estimate	Std. error	test stat.	p -value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.59	< 0.0001

Construct a 95% CI for the TV slope β_1 .

5.4 Example 1: Sales on TV Advertising 2

Step 1: read off the estimate and SE. $\hat{\beta}_1 = 0.0475$, $\hat{SE}(\hat{\beta}_1) = 0.0027$.

Step 2: critical value. For 95% confidence, $\alpha = 0.05$ and $z_{\alpha/2} = 1.96$.

Step 3: margin of error.

$$ME = 1.96 \times 0.0027 \approx 0.0053.$$

Step 4: report.

$$CI = [0.0475 - 0.0053, 0.0475 + 0.0053] = [0.0422, 0.0528].$$

Interpretation. With 95% confidence, a \$1,000 increase in TV ad spend is associated with an increase in sales of between 42 and 53 units. The interval is far from zero — evidence (anticipating §6) of a nonzero linear association between TV spending and sales.

5.5 Example 2: Starting Salary on College GPA

Setup. A career-services office regresses starting salary (in dollars) on undergraduate GPA across $n = 120$ recent graduates of a regional university. OLS gives $\hat{\beta}_1 = 5,800$ with $\hat{SE}(\hat{\beta}_1) = 950$. Construct a 95% CI for β_1 .

Step 1: read off. $\hat{\beta}_1 = 5,800$, $\hat{SE}(\hat{\beta}_1) = 950$.

Step 2: critical value. $z_{0.025} = 1.96$.

Step 3: margin of error.

$$ME = 1.96 \times 950 \approx 1,862.$$

5.5 Example 2: Starting Salary on College GPA 2

Step 4: report.

$$CI = [5,800 - 1,862, 5,800 + 1,862] \approx [\$3,938, \$7,662].$$

Interpretation. With 95% confidence, each 1.0-point increase in GPA is associated with a starting salary between \$3,938 and \$7,662 higher.

Compare with Example 1. Here the SE is about 16% of the point estimate, so the CI is relatively wide; in Example 1 the SE was about 6% of the estimate, so the CI was tight. Two knobs: smaller σ and larger X -spread make CIs narrower.

6. Hypothesis Testing for the Slope

6.1 Why the Slope Is the Question

In a simple linear regression, $\beta_1 = 0$ means

$$Y = \beta_0 + 0 \cdot X + \epsilon = \beta_0 + \epsilon.$$

X has no linear association with Y in this model. The fitted mean collapses to a constant plus noise.

So the test $H_0 : \beta_1 = 0$ is asking the question:

- Is ad spending linearly associated with sales?
- Does an additional year of schooling linearly predict earnings?
- Does the survey response *predict the outcome at all*?

This is the most-asked test in applied statistics. Output tables print its result for every coefficient by default.

6.2 Setup: Null and Alternative

Hypotheses (two-sided default)

$$H_0 : \beta_1 = 0 \quad H_1 : \beta_1 \neq 0.$$

One-sided variants exist:

- $H_1 : \beta_1 > 0$ (e.g., “the ad campaign helps”).
- $H_1 : \beta_1 < 0$ (e.g., “the policy reduces costs”).

We default to two-sided unless the problem rules out one direction. Pick the direction *before* seeing the data.

6.3 The Test Statistic

Test statistic for $H_0 : \beta_1 = 0$

$$\text{stat} = \frac{\hat{\beta}_1 - 0}{\hat{\text{SE}}(\hat{\beta}_1)} = \frac{\hat{\beta}_1}{\hat{\text{SE}}(\hat{\beta}_1)}$$

Under H_0 and the model in §3.1, for $n \geq 60$ this is approximately $N(0, 1)$.

Reading (informal; the exact cutoff comes from $z_{\alpha/2}$ in §6.4). The statistic counts how many standard errors $\hat{\beta}_1$ sits away from 0:

- $|\text{stat}|$ around 1 or less: close to 0, no reason to reject.
- $|\text{stat}|$ around 2 or more: far from 0, H_0 starts to look unlikely.

Naming aside. Regression software labels this column “t-statistic.”

Under our large-sample convention (L4 §3.3) we treat it as approximately $N(0, 1)$: same number, same decision rule.

6.4 Critical-Value Method

Decision rule (critical-value method)

Compute stat from §6.3. Reject H_0 if it falls in the rejection region:

- Two-sided ($H_1 : \beta_1 \neq 0$): $|\text{stat}| > z_{\alpha/2}$.
- Right-tailed ($H_1 : \beta_1 > 0$): $\text{stat} > z_{\alpha}$.
- Left-tailed ($H_1 : \beta_1 < 0$): $\text{stat} < -z_{\alpha}$.

Picture (two-sided, $\alpha = 0.05$):



6.5 P-Value Method

Definition — p-value

The **p-value** is the probability, under H_0 , of observing a test statistic at least as extreme as the one we got. For the two-sided test of $\beta_1 = 0$:

$$p = 2 \cdot \mathbb{P}(Z > |\text{stat}_{\text{obs}}|), \quad Z \sim N(0, 1).$$

Decision rule. Reject H_0 if $p \leq \alpha$.

Reading.

- Small p (≤ 0.05): the data we got would be very unusual if β_1 were really 0 — reject.
- Large p : the data is consistent with $\beta_1 = 0$ — fail to reject.

Software output (the last column of a regression table) reports the p-value automatically.

6.6 Recipe

How to test $H_0 : \beta_1 = 0$

1. State $H_0 : \beta_1 = 0$ and the chosen H_1 .
2. Pick a significance level α (default 0.05).
3. Compute $\text{stat} = \hat{\beta}_1 / \hat{\text{SE}}(\hat{\beta}_1)$.
4. *Either* compare $|\text{stat}|$ to $z_{\alpha/2}$ (critical-value method), *or* compute $p = 2\mathbb{P}(Z > |\text{stat}|)$ and compare to α (p-value method). The two methods always agree.
5. State the conclusion in the units of the problem.

Identical structure to L4 §7. The only thing that's new is the ingredient list: $\hat{\beta}_1$ and $\hat{\text{SE}}(\hat{\beta}_1)$ in place of $\bar{X} - \mu_0$ and $\hat{\text{SE}}(\bar{X})$.

6.7 Example 1: TV Ads on Sales (Reject)

Setup. Same regression output as §5.4. The retailer asks: does TV advertising linearly predict sales, or is the observed slope just noise? Test at $\alpha = 0.05$.

Step 1: hypotheses.

$$H_0 : \beta_1 = 0 \quad H_1 : \beta_1 \neq 0 \quad (\text{two-sided}).$$

Step 2: $\alpha = 0.05$, so $z_{\alpha/2} = 1.96$.

Step 3: test statistic.

$$\text{stat} = \frac{\hat{\beta}_1}{\widehat{\text{SE}}(\hat{\beta}_1)} = \frac{0.0475}{0.0027} \approx 17.59.$$

6.7 Example 1: TV Ads on Sales (Reject) 2

Step 4 (critical-value method). $|17.59|$ is far beyond 1.96. **Reject H_0 .**

Step 4 (p-value method, equivalent).

$$p = 2 \cdot \mathbb{P}(Z > 17.59) \approx 0 \ll 0.05.$$

Reject H_0 .

Step 5: conclusion. Overwhelming evidence of a nonzero linear association between TV ad spend and unit sales. For a \$1,000 increase in TV budget, the fitted line predicts about 47.5 additional units, and the estimate is many standard errors away from zero.

Cross-check. The 95% CI from §5.4 was $[0.0422, 0.0528]$, which excludes 0. Same conclusion — §7 makes this equivalence explicit.

6.8 Example 2: Subscriptions on Instagram Followers (Fail to Reject)

Setup. A subscription-box startup tracks $n = 80$ weekly observations. Outcome Y : new paying subscribers that week. Predictor X : growth in Instagram followers (per 100). OLS gives $\hat{\beta}_1 = 0.18$ with $\hat{SE}(\hat{\beta}_1) = 0.11$. Test at $\alpha = 0.05$ whether follower growth predicts subscription growth.

Step 1: hypotheses.

$$H_0 : \beta_1 = 0 \quad H_1 : \beta_1 \neq 0 \quad (\text{two-sided}).$$

Step 2: $\alpha = 0.05$, $z_{\alpha/2} = 1.96$.

6.8 Example 2: Subscriptions on Instagram Followers (Fail to Reject) 2

Step 3: test statistic.

$$\text{stat} = \frac{0.18}{0.11} \approx 1.636.$$

Step 4 (critical-value method). $|1.636| < 1.96$, so the statistic is *not* in the rejection region. **Fail to reject H_0 .**

Step 4 (p-value method).

$$p = 2 \cdot \mathbb{P}(Z > 1.636) \approx 0.102 > 0.05.$$

Fail to reject H_0 .

6.8 Example 2: Subscriptions on Instagram Followers (Fail to Reject) 3

Step 5: conclusion. At the 5% level, the data do *not* provide enough evidence to claim that Instagram follower growth predicts subscription growth.

Caveat

“Fail to reject” is *not* “ $\beta_1 = 0$.” With only $n = 80$ noisy weeks, the true slope could be modest and we simply lack the power to detect it. A larger sample or a less noisy outcome might reverse the conclusion.

7. The CI-HT Equivalence

7.1 Two Ways to Reach the Same Verdict

Equivalence

For the two-sided test of $H_0 : \beta_1 = 0$ at level α :

reject $H_0 \iff 0$ lies **outside** the $(1 - \alpha) \cdot 100\%$ CI for β_1 .

Why? Both procedures compare $|\hat{\beta}_1|$ to $z_{\alpha/2} \cdot \hat{SE}$:

- HT rejects iff $|\hat{\beta}_1|/\hat{SE} > z_{\alpha/2}$, i.e. $|\hat{\beta}_1| > z_{\alpha/2} \hat{SE}$.
- 0 is outside the CI iff $|\hat{\beta}_1| > z_{\alpha/2} \hat{SE}$ (same condition).

Cross-check our examples.

- §5.4 TV: 95% CI = $[0.042, 0.053]$, excludes 0 $\Rightarrow$ §6.7 rejects. ✓
- §6.8 Instagram: a 95% CI is $0.18 \pm 1.96 \cdot 0.11 = [-0.036, 0.396]$, includes 0 $\Rightarrow$ §6.8 fails to reject. ✓

8. Summary

8.1 Take Away — The Foundations

One template, two parameters

Inference for the slope reuses the L3–L4 large-sample machinery *unchanged*. Only the standard error itself is new.

- Estimator: a function of the data with a sampling distribution.
- Standard error: the spread of that distribution.
- Sampling distribution: approximately N for large n .
- CI: estimate $\pm z_{\alpha/2} \hat{SE}$.
- HT: $\text{stat} = \text{estimate}/\hat{SE}$, reject if $|\text{stat}| > z_{\alpha/2}$ or $p \leq \alpha$.

8.2 Take Away — The Key Results

OLS slope under the L7 model

- **Unbiased:** $\mathbb{E}[\hat{\beta}_1] = \beta_1$.
- **Consistent:** $\hat{\beta}_1 \xrightarrow{P} \beta_1$ as $n \rightarrow \infty$ (Lecture 6 §4.3–§4.5).
- **Sampling distribution:** $\hat{\beta}_1 \approx N(\beta_1, \text{SE}(\hat{\beta}_1)^2)$ for $n \geq 60$.
- **Standard error:** read off the *Std. error* column of the OLS output — the computer plugs in an estimate of σ and reports the result for us.
- **CI:** $\hat{\beta}_1 \pm z_{\alpha/2} \hat{\text{SE}}$.
- **HT:** $\text{stat} = \hat{\beta}_1 / \hat{\text{SE}}$; same decision rules as L4.

8.3 Where Each Idea Reappears

Forward pointers

- **Lecture 9 (multiple regression)**. The same CI and HT template applies to *each* coefficient separately. A new joint F -test asks “are *all* the slopes zero?”
- **Lecture 10 (regularization)**. Deliberately *biased* estimators (ridge, lasso) that pay a bias cost to lower variance — breaks the unbiased-is-best intuition built here.
- **Lecture 10 (bias–variance)**. The general bias–variance tradeoff in full generality.
- **Lecture 13 (resampling — bootstrap)**. A simulation-based alternative to the plug-in $\hat{SE}$ formula — useful when the formula is unwieldy or its assumptions break.